# Kyle Main

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## Senior Security Data Scientist & Detection Engineer — Elasticsearch-Native ML Threat Detection

12 years of security engineering experience, including 8+ years building Elasticsearch-native detection and analytics platforms and ML-based threat detection content — from an ES/Kibana-based next-gen SIEM at an early-stage startup to a from-scratch DOE/NNSA security data platform with a full UEBA layer on ES transforms. Deep clustering, time-series anomaly detection, and false-positive-reduction ML work, plus hands-on GenAI/LLM production tooling for security automation.

## ELASTICSEARCH & ELASTIC SECURITY DEPTH

- DOE/NNSA Security Data Integration: built an entirely new Elasticsearch-based security platform ingesting CrowdStrike, Suricata, and Zeek — including a UEBA detection layer built on ES transforms, custom Kibana dashboards, and native ES data-quality alerting.
- Trend Micro/Cysiv next-gen SIEM: wrote and tuned Elasticsearch Query DSL and native ES detection rules directly against production indexes as core detection content; built 50+ Logstash filters and deployed multiple Elastic Beats variants for log collection into ES.
- CISA CDM at DOE: data ingestion and data-quality engineering across a combined Elasticsearch/Splunk environment supporting a federal SOC.

*Full ES lifecycle across three employers: ES API, ES transforms, Query DSL, native detection rules/alerting, Beats collection, and Kibana dashboarding — not just log shipping.*

## ML & SECURITY DATA SCIENCE

**Detection & Anomaly Modeling:** Clustering/unsupervised ML for device behavior classification; time-series anomaly detection for entity behaviors (auth volume/geography, process-chain deviations); signature, statistical, behavioral, and ML-based detection rule development with formally tracked false-positive rates and staged rollout

**ML Frameworks & Languages:** Python, pandas, scikit-learn, NumPy, SciPy, PyTorch, SQL, R

**GenAI/LLM for Security:** Prompt engineering for detection triage and rule generation; GenAI-driven orchestration of SIEM APIs across 9 platforms; GenAI-powered cross-SIEM rule-conversion tooling for other engineers — hands-on LLM-API-into-production experience

## PROFESSIONAL EXPERIENCE

**Senior Cybersecurity Engineer — Shorepoint** *Oct 2023 – Present*

- Treasury SOC (TSSOC), Threat & Research team: own the detection/alerting content (Splunk saved searches) driving live SOC incident response. Previously built DOE/NNSA's Elasticsearch security platform (above) and supported CISA CDM data-quality work at DOE.

**Senior Threat Detection Engineer & Data Scientist — Forescout** *Aug 2022 – Oct 2023*

- Senior member of the data science/detection engineering team building signature, behavioral, statistical, time-series, and ML-based detection content against cloud-scale customer data; incorporated Vedere Labs threat intelligence into detection tuning.

**Threat Detection Engineer & Data Scientist — Trend Micro / Cysiv** *Sep 2018 – Aug 2022*

- Early hire building the ES/Kibana-based next-gen SIEM's rules engine and detection content — 2,300+ rules across most of the MITRE ATT&CK matrix; data engineering for 220+ log sources; GCP Dataproc/PySpark exploratory data analysis at scale.

**Security Data Scientist — Experian** *Jan 2015 – Jan 2018*

- Built DNS-based detection and mitigation for malware infections; analyzed large-scale security log data to surface anomalous behavior.

## EDUCATION & CERTIFICATIONS

M.S. Physics — University of North Texas (2013), Numerical Data Analysis & Modeling, Applied Physics | B.S. Physics — Ball State University (2011), Minors: Mathematics, Astrophysics

Splunk User Certification · Splunk for Analytics and Data Science · Johns Hopkins (Coursera): R Programming, The Data Scientist's Toolbox

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Threat Research and Detection Engineering Team
Elastic

Elasticsearch has been the core detection and analytics platform across nearly my entire career — not a peripheral tool, but the system I've built from the ground up. At DOE/NNSA I built an entirely new Elasticsearch-based security data platform from scratch, including its full UEBA detection layer; at Trend Micro/Cysiv I wrote and tuned Elasticsearch Query DSL and native detection rules as core content for an ES/Kibana-based next-gen SIEM; and at DOE I supported data ingestion and data-quality work inside a combined Elasticsearch/Splunk environment. Few candidates bring that kind of direct, cross-employer, full-lifecycle Elasticsearch depth to a Principal Security ML Research Engineer role.

Beyond Elasticsearch, my work has centered on ML-based threat detection: clustering devices on the network by behavioral features, time-series anomaly detection across authentication and process-chain behaviors, and building signature, statistical, behavioral, and ML detection rules with formally tracked false-positive rates and staged rollout before full production deployment. At Trend Micro/Cysiv, this work scaled to 2,300+ detection rules across most of the MITRE ATT&CK matrix, engineered to catch real threats while minimizing false positives for analysts downstream.

I've also put LLM APIs into production for security use cases directly — prompt engineering for detection triage and rule generation, GenAI-driven orchestration of SIEM APIs across nine platforms, and GenAI-powered tooling that converts detection rules between SIEM syntaxes for other engineers. That combination of deep Elasticsearch expertise, ML detection engineering, and hands-on production LLM integration maps directly to what this role is asking for.

I'd welcome the chance to talk through how this background could contribute to Elastic Security's threat research roadmap.

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